

PRANAV GUPTA

Dallas, TX · pranavgupta5581@gmail.com · [linkedin.com/in/pgupta55](https://www.linkedin.com/in/pgupta55) · pranavgupta.co · github.com/pranavgupta55

EDUCATION

University of Pennsylvania

Robotics MSE, Artificial Intelligence BSE with Data Science Minor

Coursework: Mathematical Foundations of Computer Science, Calculus III, Artificial Intelligence Lab

Philadelphia, PA

August 2025 - May 2029

School of Science and Engineering

SAT: 1580, GPA: 4.58

Dallas, TX

August 2021 - May 2025

TECHNICAL SKILLS

Languages: Python, Java, OCaml, JavaScript, SQL, HTML/CSS

Frameworks / Libraries: React, FastAPI, PyTorch, Pandas, LangChain, Node.js, Tailwind

AI / ML: Full-stack LLM integration, Prompt Engineering, RAG, Self & Cross Attention, PPO/DQN, CNNs, LSTMs

Tools: Git, Docker, PostgreSQL, Supabase, OpenAI API, Vercel, Fusion 360, Onshape, SolidWorks

EXPERIENCE

Autonomous Driving Research Intern

June–Sept 2024

Student Intern & First Author

University of Texas at Dallas

- Authored and published an ML research paper in Curieux Academic Journal; handled data pipeline, experiments, and technical writing using **Python**, **TensorFlow/TensorBoard**, and **Pandas**. ([view journal](#) or [view arXiv](#))
- Developed and trained 15+ models with **Keras** on 800k datapoints from the nuScenes dataset to reduce training times from 36gpu hours to 24gpu hours while maintaining model accuracy.

Co-Founder & Co-Captain

Aug 2022 – Present

National Solar Car Challenge

Dallas, TX

- Led a 9-person team to design, fundraise (\$6,500+), build, and test a one-seater solar car (45 mph top speed) to place 4th nationally in 2025 as a second year team by finding 17+ sponsors, registering the team as a 501(c)(3) non-profit, and learning/teaching others how to **weld**, **angle grind**, and **CAD** parts
- Co-designed and built chassis, crush zone, panel array, and chain-drive; CAD modeled components and validated structural performance in **Fusion 360** using stress analysis tools.

PROJECTS

Recall AI · Full-stack AI Study Platform

Jul–Aug 2025

- Built a FastAPI + GPT-4o tutoring chatbot with token-saving UX (localStorage + message summaries) and LangChain enforcement of structured JSON outputs.
- Implemented a cache-then-database sync (PostgreSQL) and Row Level Security to enforce per-user data isolation.

Python Coding Projects

Jun 2021 – Present

- Created 100+ Python projects (AI simulations, physics engines, procedural generation) with Pygame UIs.
- Trained a PPO RL agent (PyTorch) for drone control in a custom 2D physics simulator; implemented OpenGL visual-effect renderers (shaders, Delaunay, ray-casting); built particle-based fluid dynamics, cloth physics, and soft-body physics simulations

LEADERSHIP & ACTIVITIES

Current Penn Clubs

University of Pennsylvania

- PennAiR Software Division, AI@Penn Research and Development Divisions, AIBC Development Division

Captain, Design Team

National Solar Car Challenge

- Led Fusion 360 CAD workshops; mentored members in modeling, tolerancing and manufacturing prep.

Co-Founder & Captain

Rocketry Club, SEM High School

- Organized lessons and launches for 12+ model rockets; taught aerodynamics, stability and basic electronics.

Eagle Scout

BSA Troop 840

- Led a year-long community project raising \$1,500+ to build a playground for a local elementary school; completed 24 merit badges.

AWARDS & HONORS

- National Merit Scholarship Finalist; 3× UIL State Champion (Math & Science).
- Co-VP of VEX Robotics — 2nd place regional; Member of a FIRST Robotics World-Championship team
- Completed 16 AP exams with 14 scores of 5 (Calculus BC, Physics C, Computer Science A among them).